Engineering Clinic:

Review -2

Contretemps Nullification

Welcome to our presentation on Contretemps Nullification. This project aims to mitigate unexpected events. We will explore the design, components, and functionality. Our team will guide you through the system.

System Diagram:

Schematic

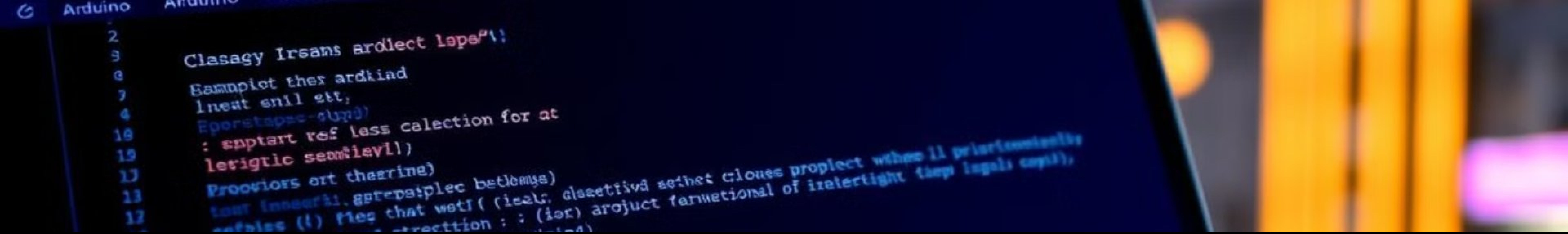
Overall design of the system.

Code

Logic and functionality.

Component

Physical elements.



Code

```
// Define pins for the first ultrasonic sensor
const int trigPin1 = 18;
const int echoPin1 = 21;

// Define pins for the second ultrasonic sensor
const int trigPin2 = 19;
const int echoPin2 = 22;

// Define pins for the buzzers
const int buzzer1 = 23;
const int buzzer2 = 5;

// Define distance threshold in cm
const int distanceThreshold = 20;

void setup() {
  // Initialize serial monitor
```

```
// Define pins for the first ultrasonic sensor
const int
  // Initialize serial monitor
  Serial.begin(115200);

  // Set pins for ultrasonic sensors
  pinMode(trigPin1, OUTPUT);
  pinMode(echoPin1, INPUT);
  pinMode(trigPin2, OUTPUT);
  pinMode(echoPin2, INPUT);

  // Set pins for buzzers
  pinMode(buzzer1, OUTPUT);
  pinMode(buzzer2, OUTPUT);
}

void loop() {
```

```
// Measure distance for the first sensor
long distance1 = measureDistance(trigPin1, echoPin1);

// Measure distance for the second sensor
long distance2 = measureDistance(trigPin2, echoPin2);

// Print distances to serial monitor
Serial.print("Distance 1: ");
Serial.print(distance1);
Serial.print(" cm, Distance 2: ");
Serial.print(distance2);
Serial.println(" cm");

// Check distance threshold for the first sensor
if (distance1 <= distanceThreshold) {
  digitalWrite(buzzer1, HIGH);  // Turn on buzzer 1
} else {
```

```
// Check distance threshold for the first sensor
if (distance1 <= distanceThreshold) {
digitalWrite(buzzer1, HIGH);  // Turn on buzzer 1
} else {
digitalWrite(buzzer1, LOW);   // Turn off buzzer 1
}

// Check distance threshold for the second sensor
if (distance2 <= distanceThreshold) {
digitalWrite(buzzer2, HIGH);  // Turn on buzzer 2
} else {
digitalWrite(buzzer2, LOW);   // Turn off buzzer 2
}

delay(100);  // Delay to avoid excessive measurements
}

// Function to measure distance using ultrasonic sensor
long measureDistance(int trigPin, int echoPin) {
digitalWrite(trigPin, LOW);
delayMicroseconds(2);
digitalWrite(trigPin, HIGH);
delayMicroseconds(10);
digitalWrite(trigPin, LOW);

long duration = pulseIn(echoPin, HIGH);
long distance = duration * 0.034 / 2;  // Convert to distance in
cm
return distance;
}
```

Core Components:

ESP32

Microcontroller for processing.

Ultrasonic Sensor

Detects object distance.

Buzzer & LED

Provides alerts.

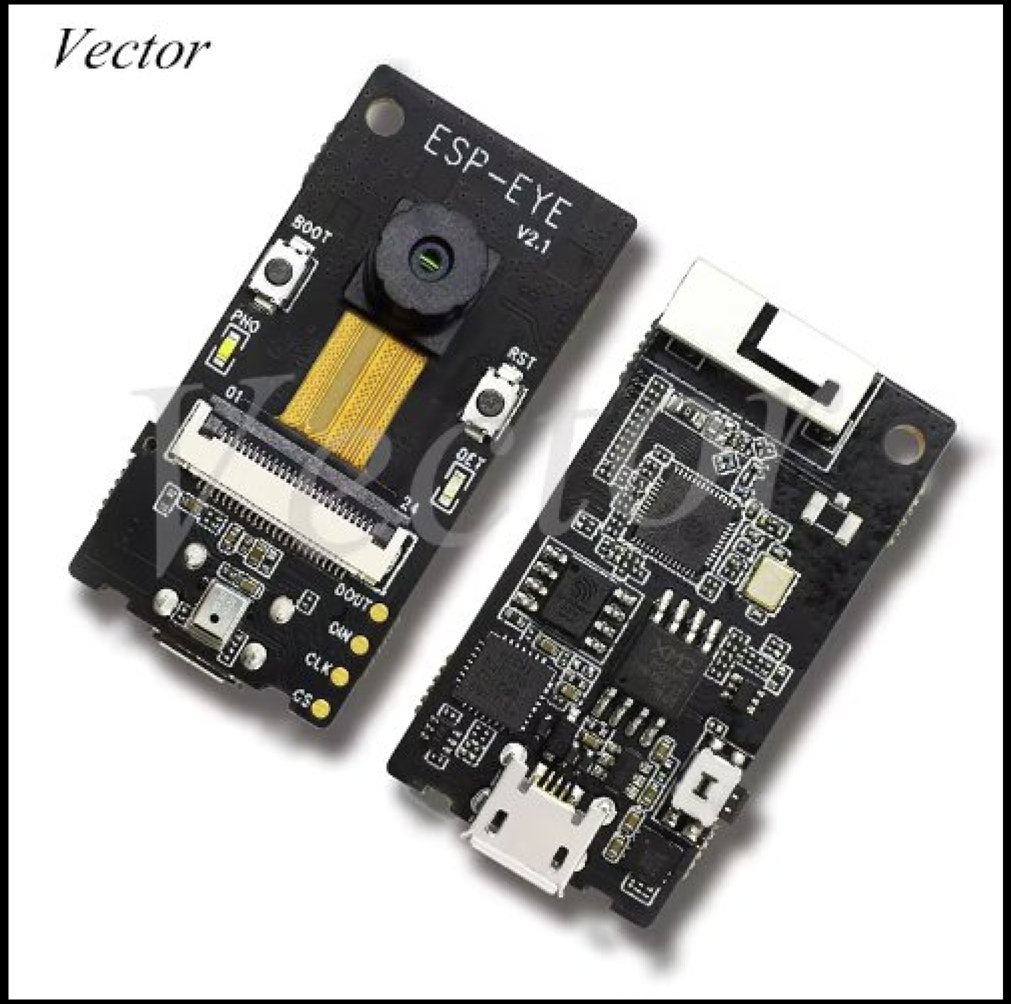
Power Source

Provides electricity.

ESP32 Microcontroller:

The ESP32 integrates a microcontroller. It is coupled with software for coding. Code is written on a computer. It is then uploaded to the board.

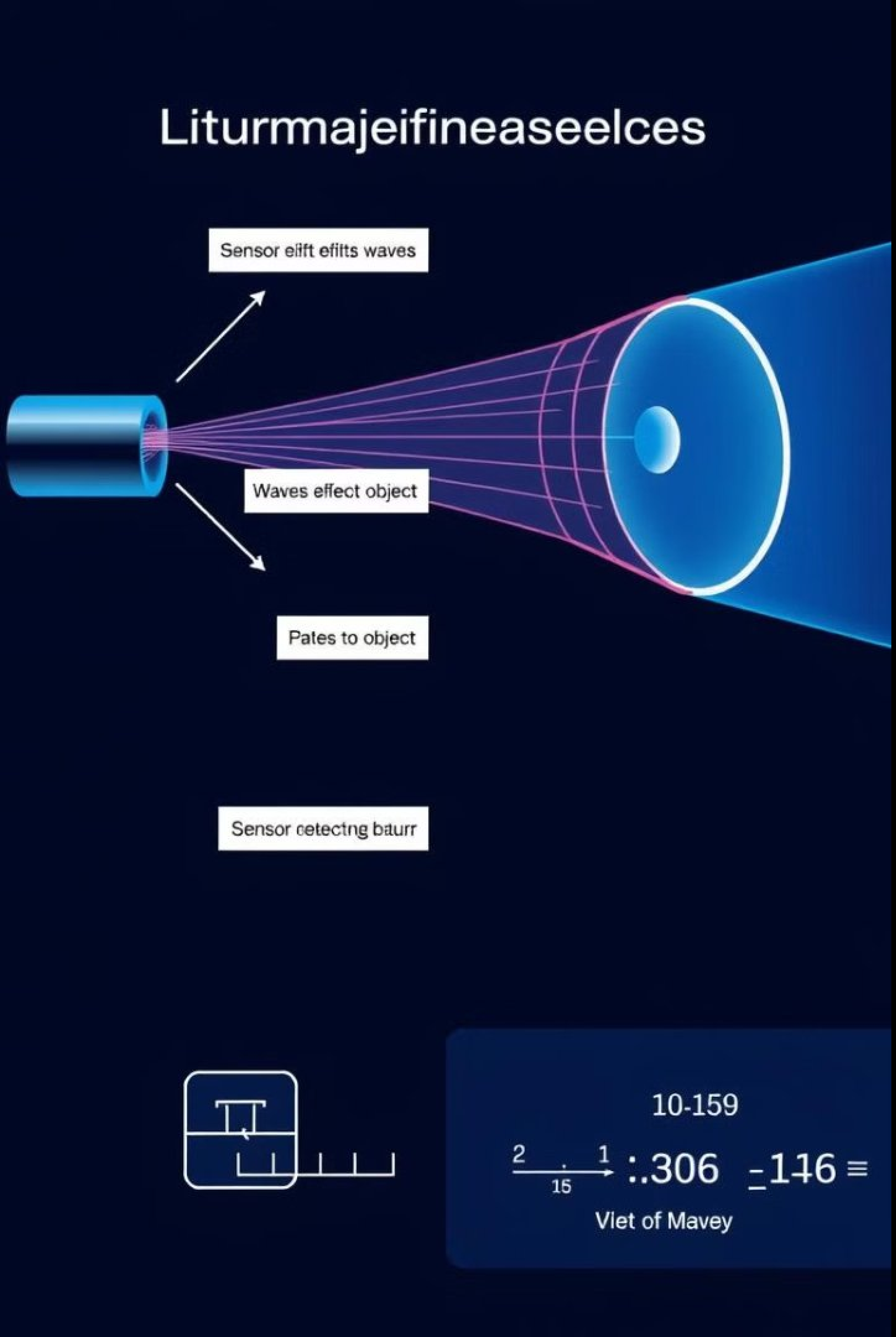
It's a versatile tool. It's suited for various applications. Its processing power is substantial.



Breadboard and Wiring:

Breadboards create circuits temporarily. They are solderless. This makes them reusable.

Connecting wires provide pathways. Electricity flows between components. Proper wiring is vital. It ensures circuit operations



Ultrasonic Sensor Details

1 **Transmission**

Emits sound waves.

2 **Reflection**

Waves bounce off objects.

3 **Detection**

Sensor detects return.

4 **Calculation**

Distance determined by time.

*Alert System: Buzzer and LED:

The buzzer and LED provide alerts. They receive signals. The signals come from the sensor. This notifies the driver. This is triggered when an object is detected.

It indicates a potential collision. It also indicates other issues. The system helps the driver. It enhances safety measures

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Thank you